III B.Tech II Semester

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23A39605	EDGE COMPUTING **LAB** **(Professional Core)**	**0**	**0**	**3**	**1.5**

Course Objectives:

- To impart practical knowledge of edge computing architecture and frameworks.
- To provide hands-on experience in deploying AI and IoT applications on edge devices.
- To explore edge-native computing environments like Raspberry Pi, Jetson Nano, etc.
- To introduce performance analysis, optimization, and energy efficiency in edge computing.
- To understand data preprocessing, model deployment, and real-time inferencing on the edge.

Course Outcomes:

- Students will be able to practically implement edge computing applications using real-world platforms and tools.
- Students gain experience in deploying lightweight ML models on edge devices, manage data flow between edge and cloud, and ensure system performance under limited computing and networking resources.
- Students will understand how to optimize latency, power consumption, and reliability of edge solutions for smart environments.

List of Lab Experiments:

1. **Setup and Configuration of Edge Devices**
   - Raspberry Pi/Jetson Nano installation, SSH, GPIO control
2. **Data Acquisition from Sensors**
   - Reading data from temperature, motion, and environmental sensors
3. **Deploying a Lightweight ML Model on Edge**
   - Using TensorFlow Lite or PyTorch Mobile for deployment
4. **Real-Time Image Classification at the Edge**
   - Using camera module with edge device for inference
5. **MQTT-Based Edge Communication**
   - Setup publisher/subscriber model for edge-to-cloud communication
6. **Integrating Edge Devices with Cloud Platforms**
   - AWS IoT, Azure IoT Hub, or Google Cloud IoT integration
7. **Edge Device Power and Latency Monitoring**
   - Measuring latency and energy consumption during model inference
8. **Edge AI Application – Smart Surveillance System**
   - Face or object detection on live video stream using OpenCV
9. **Streaming Data Analytics on Edge**
   - Local aggregation and event processing with Kafka or lightweight alternatives
10. **Model Optimization for Edge Deployment**

11. Quantization, pruning, and compression for reducing model size
    TinyML for Microcontroller-Based Inference

12. Deploy a model on Arduino/Nano BLE using TensorFlow Lite Micro
    Edge-Orchestrated Federated Learning Prototype

- Basic FL setup using two edge devices sharing a model

Text Books:

1. Perry Lea, *Edge Computing: From Hype to Reality*, Manning Publications.
2. Satyanarayana G. et al., *Edge Analytics with Raspberry Pi*, Springer.
3. Arun Kumar Sangaiah et al., *Edge Computing and Computational Intelligence Paradigms for the IoT*, Elsevier.

Reference Books:

1. Flavio Bonomi, *Fog and Edge Computing: Principles and Paradigms*, Wiley.
2. Preetha Evangeline, *IoT and Edge Computing for Architects*, Packt Publishing.
3. Rajkumar Buyya, *Internet of Things and Edge Computing for Smart Environments*, Springer.

Online Courses:

1. Edge AI and Computer Vision – Udacity (NVIDIA)
2. **AI on the Edge with Raspberry Pi – Coursera**